

Pilot Summary - Test-Time Canonicalization for Robust VLA

Summary: 2-Day Pilot Results and Proposal

Date: 2026-09-02 · **Machine:** Blackwell RTX PRO 6000 (96 GB) · **Location:** /home/user/Desktop/novelty_module/

Running now (unattended overnight)

Job	Checkpoint	Suite	Progress	Out
DeepONet 4-arm pilot	don_v2_30k_noaug_s0 (30K)	Spatial	20/96 rollouts	results/pilot_c
Flow Goal pilot (chained)	flow8300_goal_s0	Goal	queued	results/pilot_f
Flow Object pilot (chained)	flow8300_object_s0	Object	queued	results/pilot_f
Flow Long pilot (chained)	flow8300_10_s0	Long	queued	results/pilot_f

ETA: ~7-8 h total (DeepONet first, then 3 flow suites sequentially to avoid GPU contention). Logs: pilot_deeponet.log, pilot_flow_goal.log, pilot_flow_object.log, pilot_flow_10.log, chain.log.

What was done today

- 1. **Flow pilot (done):** 1-seed paired 4-arm pilot on LIBERO-Spatial — see table below.

- 2. **Harness extended:** eval_canon_pilot.py --head deeponet — auto-resolves the lab’s training-provenance guard from run_config.json (verified: head=deeponet, bandlimit=0, pool_norm=0, P=256, blocks=3, queries=8, fourier=16). Same protocol, identical tasks + init states across both backbones.
- 3. **Proposal updated:** two-axis scaled study (cross-suite on flow, cross-backbone on Spatial).

Flow pilot results (Spatial, 1 seed, directional)

Arm	Light	Background	Objects	Camera	Avg
base	0.500	0.167	0.167	0.167	0.25
+canon	0.667	0.167	0.167	0.333	0.33
+consensus	0.667	0.167	0.167	0.500	0.37
+canon+consensus	0.333	0.167	0.167	0.333	0.25

Key findings

- 1. **Canonicalization hits its target:** +16.7pp Light, +16.7pp Camera. Zero on Background/Objects (expected — illumination module, not texture/layout).
- 2. **Consensus is broader:** +33.3pp Camera, +16.7pp Light (noise-averaging handles viewpoint jitter).
- 3. **Naive composition hurts** (canon+consensus 0.333 vs consensus 0.667 on Light) — the interference is the finding; motivates a learned composition gate.
- 4. **Background/Objects need architecture-level solutions** — exactly what the DeepONet pilot (running) will test.

Module properties (reviewer-facing)

Property	Canon	Consensus
Parameters	0	0
Training required	none	none
Plug-and-play	yes (obs pre-transform)	yes (N forwards + mean)
Overhead	µs-scale	N× forwards at replan (~+3.4% wall-clock at N=4, replan=5)

The proposal

Positioning: *“While recent modules fix single perturbation families (AnyCamVLA: viewpoint; PDF: object pose), we give the first test-time module for illumination/texture shift and show it composes with architecture-level robustness, compounding gains on LIBERO-Plus.”*

Scaled study (two axes, all checkpoints already on disk): 1.

Cross-suite (flow, 4 suites): Spatial/Goal/Object/Long $\times$ {base, +canon, +consensus} $\times$ 5 seeds. 2. **Cross-backbone (Spatial):** flow8300 vs DeepONet-30K vs DeepONet-v2 $\times$ 4 arms — directly answers “does it compose with DeepONet?” and “is DeepONet already covering Light/Camera?”. 3. Learned composition gate (fixes the pilot’s interference finding). 4. Ablations: gain steps, $N \in \{2,4,8\}$, σ sweep, latency.

Venues: NeurIPS/ICML 2027 (primary) · CoRL/IROS 2027 · RA-L (Q2 safety net).

Honest caveats

- 1-seed deltas are directional only (lab’s documented 10.5-21.9% rerun noise floor); paired design cancels to first order; 5-seed study is confirmatory.
- DeepONet comparison is Spatial-only (only checkpoint on disk).
- Background/Objects: input-side modules don’t touch them — this is stated, not hidden.